Primary Research papers for the project

For your Fairness and Bias Audit project, which focuses on the **diagnostic evaluation** of facial analysis systems and alignment with regulatory frameworks like the **EU AI Act**, here are 10 highly relevant and recent research papers (published 2024–2026) that align with your core objectives:

1. Fairness and Bias Mitigation in Computer Vision: A Survey (2024)

- **Core Alignment:** This is the most recent comprehensive survey of your specific domain. It catalogues the sources of bias across the machine learning pipeline and provides a family of metrics and benchmarks essential for a diagnostic tool.
- **Key Value:** It validates your project's focus by highlighting the comparative scarcity of repeatable auditing pipelines in vision research.

2. Assessing High-Risk AI Systems under the EU AI Act (2026)

- **Core Alignment:** This paper provides a structured mapping that translates high-level legal requirements into **concrete, implementable verification activities**.
- **Key Value:** It directly informs your "Regulatory Traceability" goals by decomposing abstract legal obligations into auditable technical controls.

3. Where, Why, and How is Bias Learned in Medical Image Analysis Models? (2024)

- **Core Alignment:** A study on how bias is encoded within convolutional networks even without explicit demographic labels.
- **Key Value:** It supports your project's diagnostic mission by demonstrating that auditing tools must detect disparities not always attributable to obvious single causes.

4. Automatic Generation of Model and Data Cards: A Step Towards Responsible AI (2024)

- **Core Alignment:** Focuses on automating the generation of the exact reporting artifacts your project intends to deliver.
- **Key Value:** It explores using retrieval-based methods to populate documentation fields, which could inform your **Report Generation Module**.

5. What's Documented in AI? Systematic Analysis of 32K AI Model Cards (2024)

- **Core Alignment:** An empirical assessment of real-world documentation practices on Hugging Face.

- **Key Value:** It identifies common underreporting gaps in evaluation and limitation sections, helping you design a more robust report template that avoids these "common failure modes".

6. An Open Knowledge Graph-Based Approach for Mapping Concepts between the EU AI Act and International Standards (2025)

- **Core Alignment:** Bridges the gap between statutory requirements and technical standards like ISO/IEC and NIST.
- **Key Value:** It provides a machine-readable way to ensure your audit metrics align with the "Core Four" regulatory expectations.

7. MRM3: Machine Readable ML Model Metadata (2025)

- **Core Alignment:** Proposes compute-friendly representations for model metadata.
- **Key Value:** This aligns with your goal of creating standardized, exportable reports (HTML/PDF) by providing an ontological structure for your internal schema.

8. From AI Act to Structured Testing of AI Systems (2024)

- **Core Alignment:** Specifically focuses on the transition from regulatory text to **structured testing protocols**.
- **Key Value:** It supports your acceptance criteria (like $\alpha=0.05$ and bootstrap resamples) by grounding them in emerging European testing norms.

9. Laminator: Verifiable ML Property Cards using Hardware-Assisted Attestations (2024)

- **Core Alignment:** Research into ensuring the integrity and verifiability of reported metrics.
- **Key Value:** While advanced, it addresses the "Ethical Consideration" of trust in auditing results, ensuring that a diagnostic report hasn't been tampered with.

10. FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age (2021/Revised 2024)

- **Core Alignment:** The foundational paper for one of your core datasets and your minimum case study model (ResNet-34).
- **Key Value:** Essential for understanding the taxonomy and demographic balance logic that your **Data Ingestion Module** must handle.

These papers emphasize the shift from ad-hoc evaluations to **structured, repeatable, and regulatory-aligned auditing pipelines**, mirroring the core engineering strategy of your project

Secondary Research Papers for the Project

To assist with your specific engineering modules (Metrics Engine, Data Ingestion, and Orchestration), here are 10 **Secondary Research Papers** that provide deeper technical and operational guidance:

1. Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing (2020/Updated 2022)

- **Module Assistance: Orchestration and Configuration Layer**.
- **Why it helps:** Cited in your feasibility study, this paper provides the organizational workflow for moving from a "one-off" audit to an integrated "internal audit" process within an engineering team.

2. The four-fifths rule is not disparate impact: a woeful tale of epistemic trespassing in algorithmic fairness (2022)

- **Module Assistance: Fairness Metrics Engine**.
- **Why it helps:** It provides a critical warning against misapplying the "80% rule" (Demographic Parity) and explains the statistical dangers of "epistemic trespassing" when interpreting your "Core Four" metrics.

3. Pervasive label errors in test sets destabilize machine learning benchmarks (Northcutt et al., 2021)

- **Module Assistance: Data Ingestion (Risk Management)**.
- **Why it helps:** Directly addresses your flagged risk of **"label noise"** in benchmarks like UTKFace. It offers a framework for understanding how erroneous labels in your test matrix can distort your fairness findings.

4. Equality of Opportunity in Supervised Learning (Hardt et al., 2016)

- **Module Assistance: Fairness Metrics Engine**.
- **Why it helps:** This is the foundational mathematical source for the **Equal Opportunity** and **Equalised Odds** metrics you are required to compute.

5. Addressing Inclusion in Face Recognition (Paravision White Paper, 2026)

- **Module Assistance: Report Generation (Case Study)**.
- **Why it helps:** Provides the most recent (mid-2026) industry baseline for demographic error differentials (FMR Max). You can use its findings to contextualize the ResNet-34 audit results in your final report.

6. Model Reporting for Certifiable AI: A Proposal from Merging EU Regulation into AI Development (2023)

- **Module Assistance: Report Generation (Regulatory Traceability)**.
- **Why it helps:** It proposes "Regulatory Cards" that unify data, model, and operational requirements into one bundle—directly supporting your goal of **Regulatory Traceability**.

7. On the legal compatibility of fairness definitions (Xiang and Raji, 2019)

- **Module Assistance: Legal and Ethical Feasibility**.
- **Why it helps:** Essential for your "Legal Feasibility" chapter. It explores how mathematical fairness definitions collide with legal concepts of discrimination, helping you justify your choice of metrics to a non-technical auditor.

8. Image representations learned with unsupervised pre-training contain human-like biases (Steed and Caliskan, 2021)

- **Module Assistance: Fairness Metrics Engine**.
- **Why it helps:** Supports your diagnostic mission by demonstrating that bias is encoded in model "embeddings" even when explicit demographic labels are absent from the training data.

9. Data Cards: Purposeful and Transparent Dataset Documentation for Responsible AI (2022)

- **Module Assistance: Report Generation (Datasheets)**.
- **Why it helps:** It refines the "Datasheets for Datasets" concept into modular, user-centric summaries (telescopic, periscopic, microscopic) that can inform your **Report Generation Module**'s layout.

10. Review of Demographic Bias in Face Recognition (arXiv, 2025)

- **Module Assistance: Project Literature Review**.
- **Why it helps:** A more recent domain-specific survey that updates the findings of the earlier *Gender Shades* study with 2025 benchmark data from NIST and other sources.

These secondary papers will provide the **"methodological defense"** needed to support the engineering decisions you make in your five-module architecture.